Advanced Bio-Electronics

EMV - Eulerian Video Magnification

SHAHAR PERETZ & SHANI ASRAF

Department of Biomedical Engineering, Faculty of Engineering, Tel Aviv University, Tel Aviv, Israel.
Corresponding authors: shaharperetz@mail.tau.ac.il , shaniasraf@mail.tau.ac.il

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Abstract

EMV approaches amplifies the actual motion and allows us to recognize movements that are undetectable to the naked eye. The basic methodology of EMV considers the time series of pixels and amplifies any variation in a specified interest band of temporal frequency. EMV may also have significant potential in diagnostic and monitoring in medical applications [1]. The first part of the experiment includes a comparison and an analysis between two methods of measuring Heart Rate (HR): EMV and ECG, when the EMV algorithm has been used to amplify subtle changes in color changes. From the first part of the experiment, it could be concluded that the EMV can successfully evaluate HR when taken while the examinee is at rest. However, less accurate results were obtained from the recording that was taken after exercise, but since the results weren't too far from being accurate enough, it can be assumed that some adaptations and new requirements in the experiment procedure can improve the accuracy of the results that were taken after exercising as well. Second part of the experiment includes a comparison and an analysis between two methods of measuring Respiratory Rate (RR) while normal breathing (eupnea): EMV and respiration test, when the EMV algorithm has been used to amplify subtle changes in movement changes. It could be concluded that the EMV can evaluate RR when taken while the examinee is at rest while the experimental conditions maintain more uniform and with some adaptations.

1. INTRODUCTION

HR (Heart Rate)

Heart rate (HR) is the number of heartbeats per unit of time, usually per minute. The heart rate is based on the number of contractions of the ventricles (the lower chambers of the heart). The heart rate may be too fast (tachycardia) or too slow (bradycardia). The pulse is a bulge of an artery from waves of blood that course through the blood vessels each time the heart beats [2]. The HR response to exercise provides useful information about the autonomic function and has prognostic value [3].

RR (Respiratory Rate)

The respiration rate is the number of breaths a person takes per minute. The rate is usually measured when a person is at rest and simply involves counting the number of breaths for one minute by counting how many times the chest rises. Normal respiration rates for an adult person at rest range from 12 to 16 breaths per minute [4]. Respiratory rates change based on many health and activity factors such as: resting, exercising, panicking, eating, and even illness [5]. Monitoring of respiratory rate (RR) is one of the vital parameters in many areas including daily activities, sports and medical applications [6].

EMV

EMV is a non-invasive and a non-contact procedure, which amplifies the actual motion and allows us to recognize movements that are undetectable to the naked eye [1]. EMV can use the fact that human skin color varies slightly with blood circulation to measure HR, by revealing temporal variations in videos that are difficult or impossible to see with the naked eye and display them in an indicative manner [7].

The basic approach is to consider the time series of color values at any spatial location (pixel) and amplify variation in a given temporal frequency band of interest. The amplification reveals the variation of redness as blood flows through the face. For this application, temporal filtering needs to be applied to lower spatial frequencies (spatial pooling) to allow such a subtle input signal to rise above the camera sensor and quantization noise [7].

The approach combines spatial and temporal processing to emphasize subtle temporal changes in a video. The process is illustrated in Figure 1. First the video decomposes the sequence into different spatial frequency bands. These bands might be magnified differently because (a) they might exhibit different signal-to-noise ratios or (b) they might contain spatial frequencies for which the linear approximation used in our motion magnification does not hold. In the latter case, the amplification is reduced for these bands to suppress artifacts. When the goal of spatial processing is simply to increase temporal signal-to-noise ratio by pooling multiple pixels, by spatially low-pass filter the frames of the video and down sample

them for computational efficiency. All that based on computing a full Laplacian pyramid [7].

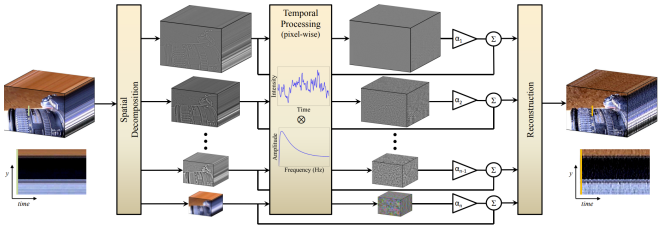

Figure 1: Overview of the Eulerian video magnification framework [7]

The following step is to perform temporal processing on each spatial band, by considering the time series corresponding to the value of a pixel in a frequency band and applying a bandpass filter to extract the frequency bands of interest, when the low and high cutoff frequencies are depending on the signal needed to amplify. After that, the extracted bandpass signal is multiplied by a magnification specific factor α, which is used to amplify the input signal. This parameter is determined based on the results desired, and the input video and it is initiated at λc - spatial frequency cutoff. At the last step, the magnified signal adds to the original and collapses the spatial pyramid to obtain the final output [7].

Remote and non-contact breathing monitoring has an increasing research interest in clinical and bio-medical applications [8]. The Eulerian motion magnification technique can also be used for measuring RR as measuring HR [9].

ECG (Electrocardiogram)

The electrocardiogram (ECG) is a non-invasive diagnostic test that evaluates the heart's electrical system to assess for heart disease. It uses flat metal electrodes placed on your chest to detect the electrical activity of the heart as it beats, which is then graphed [10]. The signal recorded is determined by action potentials of the heart generated by millions of individual cells and their sequence of activation [11].

The ECG is one of the most used tests in medicine because it can screen for a large variety of cardiac conditions, the machines are readily available in most medical facilities, the test is simple to perform, safe, and relatively inexpensive [10].

If the heart is beating steadily, it will produce the typical ECG pattern (figure 2): The first peak (P wave) shows how the electrical impulse (excitation) spreads across the two atria of the heart. The atria contract (squeeze), pumping blood into the ventricles, and then immediately relax. The electrical impulse then reaches the ventricles. This can be seen in the Q, R and S waves of the ECG, which is called the QRS complex. The ventricles contract. Then the T wave shows that the electrical impulse has stopped spreading, and the ventricles relax once again [12].

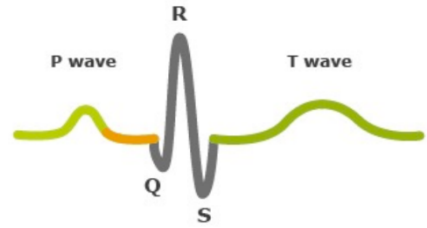

Figure 2: Course of the QRS complex in a normal heartbeat [12]

The normal duration (interval) of the QRS complex is between 0.08 and 0.10 seconds, when the duration is between 0.10 and 0.12 seconds, it is intermediate or slightly prolonged. A QRS duration of greater than 0.12 seconds is considered abnormal [13].

Objectives

The experiment includes two parts, which both have one main purpose - learning about automated tools for image processing in the medical aspect. In the first part, EMV is used for measuring HR and compared to ECG. In the second part, EMV is used for measuring RR and compared to respiration test.

2. METHODS

2.1. Part 1

2.1.1 Equipment & Materials

- camera (iPhon SE, with resolution 1080 HD at 30fps)
- ECG disposable electrodes
- Biopac system for ECG (Sampling rate = 1kHz)
- Cables
- Lab computer with Biopac student lab app 4.7v.
- MATLAB software

2.1.2. Experimental Procedure

To the examinee (female, 25 years old) were affixed three ECG patches as follows: one on right leg, second on left leg, and the last on the right wrist. The first step was recording an ECG measurement, at the same time a camera filmed and recorded a video of the examinee's face, for 30 seconds. The camera was held by a tripod in around 1.5 meters from the examinee. During this measurement the examinee was seated still when her eyes open and behind her was white background. Between the first and the second recorded, the examinee performed strenuous physical activity in the form of climbing stairs and jumping in place. Right afterwards, the measurement of the first part was taken again.

2.1.3. Data Analysis and Values Calculations

The Heart Rate (HR) calculated according to the equation:

$$(1)\quad HR = \frac{60}{\Delta t}\ [\frac{beats}{minute}]$$

Where Δt is the time difference between two adjacent peaks of the signal.

In order to compare the HR values that calculated in the different methods; EMV and ECG, a relative error value was calculated according to the equation:

$$(2)\quad \text{Relative } Error = 100 \cdot \left| \frac{x_{ECG} - x_{EMV}}{x_{ECG}} \right| \ [\%]$$

In addition, the sampling frequency given by the ECG measurement was Fs = 1000 Hz, and the sampling frequency given by the EMV method was Fs = 30 Hz, so the sampling time (Ts) was calculated according to the equation:

$$(3)\quad Ts = \frac{1}{Fs}\ [sec].$$

The mean value was calculated using the equation:

$$(4)\quad mean = \frac{1}{N}\sum_{i=1}^{N} \quad x_i$$

Where N is the number of objects in the sample; x_i is the value of object i in the sample.

The variance value was calculated using the equation:

$$(5)\quad \sigma^2 = \frac{\sum_{i=1}^{N} \ (x_i - \underline{x})^2}{N-1}$$

Where N is the number of objects in the sample; x_i is the value of object i in the sample; $\underline{x}$ is the mean value of the sample.

2.1.4. Demonstrating the Areas of interest

The area of interest in each video was determined using the MATLAB built-in feature called "ROI" which enables the user to mark the area on a picture and then extract the position of this area in the frame. By this way the same area will be retained for all the frames in the whole recording of the video.

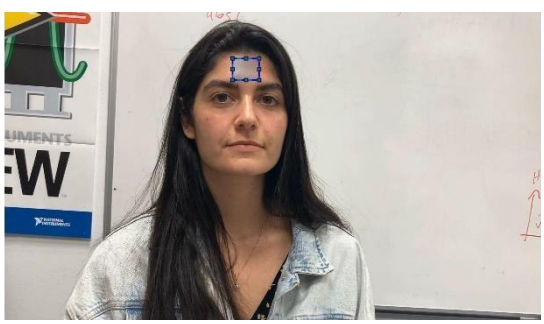

Figure 3: The region of interest at rest video

Figure 4: The region of interest at activity video

2.1.5. Extracting HR from Eulerian Video Magnification

The HR from the EMV extracted after each of the videos was cut into 30 seconds, formatted to mp4 format, and then displayed separately each of the Red, Green and Blue values from the frames as continuous signals (Appendix A). After the RGB signals displayed, the ones who had the highest power range were chosen and the HR was calculated based on the signal of the color that was chosen. The parameters that were chosen for the magnified videos are:

Table 1: The EMV Parameters that were chosen to extricate HR

	rest	**Activity**
color signal (Appendix A)	Green	Blue
f_l	60/60 Hz	113/60 Hz
f_h	70/60 Hz	133/60 Hz
α	100	80
Level	4	4
Chrom-attention	1	1

In order to extract the values of ω_l and ω_h from the values of f_l and f_h respectively there is a need to use formula (6):

$$(6)\quad \omega \ = \ 2\pi f \ [\frac{rad}{sec}]$$

Due to the calculation by formula (6) the values at rest were $\omega_l = 2\pi\ [\frac{rad}{sec}]$ and $\omega_h = \frac{7}{3}\pi\ [\frac{rad}{sec}]$. After exercise the values were $\omega_l = \frac{113}{30}\pi\ [\frac{rad}{sec}]$ and $\omega_h = \frac{133}{30}\pi\ [\frac{rad}{sec}]$. The values of the parameters "level" and "chrom-attention" were chosen due to the values in the example video in the EMV algorithm website, where there was a man with the same face color as the color of the examinee in this experiment. The ω_l and ω_h parameters were chosen due to the HRs that were calculated in the ECG measurement with a range of around 5-10 bpm from the actual HR value that was revived. In the end, the α the parameter was chosen due to the video and the graph that compared the HR in the EMV to the graph of the HR in the ECG. The goal in the selection of the parameters values was to get the HR that would be as close as possible to the HR of the ECG signals, within the assumption that the ECG signal is a more accurate measurement for HR then the EMV.

In order to minimize the effect of noises that were found in the video recording a FIR band pass filter (BPF) was applied using the built-in MATLAB functions: "designfilt" which enable the creation of the wanted filter with order 100 and the cutoff frequencies: f1=0.00000001 Hz and f2=0.9 Hz for the resting record and f1=0.00000001 Hz and f2=2.2 Hz for the activity record. "filtfilt" function was used for the sake of reducing the time delay that is typical for FIR filters.

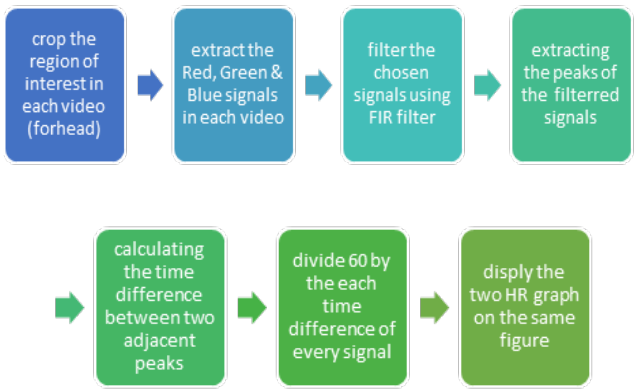

Figure 5: Block diagram of the algorithm for finding the HR from the magnified video

2.1.6. HR Extraction from ECG Measurement

The HR calculated from the ECG measurement was based on the formula (1) where Δt is extracted using the MATLAB built-in "findpeaks" function with 'MinPeakDistance' of 600 for the resting measurement and 350 for the activity measurement. Then, 60 was divided by every Δt in the sequence of each ECG signal and the HR vector was received and then displayed on a plot.

2.1.7. Compression between the HR which extracted

The comparison between the two HR signals that were calculated by the different measurements was based on relative error between the average HR calculation and by the t-test statistical hypothesis assuming normal distribution of the samples, in order to examine the significance of the results' variance. The t-test calculation is performed according to the MATLAB built-in function "ttest" comparing the average HR in the ECG and the EMV signals at rest and after exercise.

2.2. Part 2

The second part of the experiment was performed using a breast belt with a pneumograph transducer which measures the changes in force applied on the device as the breast expands and shrinks while breathing. Simultaneously, a thermistor transducer was used next to the examinee's nostril to measure the changes in temperature. This experimental part measured normal breathing (eupnea) [14].

2.2.1 Equipment & Materials

For the second part of the experiment:

- BIOPAC Respiratory Transducer SS5LB
- BIOPAC Temperature Transducer SS6L
- BIOPAC single sided (surgical) tape (TAPE1)
- Chair without armrests
- BIOPAC Student Lab System: BSL4 software, MP36
- Computer System
- MATLAB software

2.2.2. Experimental Procedure

First, the Respiratory and the temperature devices were plugged into channels 1 and 2 respectively. Then, the respiratory transducer adjusted around the examinee's chest, below the armpits and above the nipples. Next, the temperature transducer was held under the examinee's nostril (figure 6) and then the calibration procedure began. First, recording while the examinee seated and breathing normally though his nose for 8 seconds. Both channels should show graphs which behave similar to the expectation. The examinee performed normal resting breathing for 20 seconds [14].

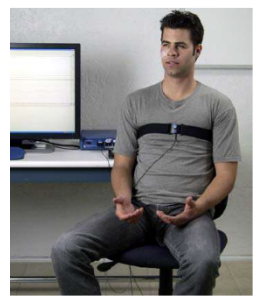

Figure 6: position of the examinee in the second part [14]

2.2.3. Data Analysis and Values Calculations

The sampling time given by the Biopac was Ts = 0.01 sec, so the sampling frequency (Fs) was calculated according to the equation:

$$(7)\quad Fs = \frac{1}{Ts}\ [Hz].$$

The mean value was calculated using the formula (4).

The variance was calculated using formula (5).

The Respiration Rate (RR) calculated according to the equation:

$$(8)\quad RR = \frac{60}{\Delta t}\ [\frac{BREATHS}{MINUTE}]$$

Where Δt is the time difference between two adjacent peaks of the respiration signal.

2.2.4. Demonstrating the Areas of interest

The area of interest in the video was determined in the same procedure which is described at section 2.1.4.

2.2.5. Extracting RR from Eulerian Video Magnification

The RR from the EMV extracted procedure which is described at section 2.1.5. The parameters that were chosen for the magnified video are:

Table 2: The EMV Parameters that were chosen to extricate RR

color signal (Appendix B)	Values
f_l	15/60 Hz
f_h	17/60 Hz
α	100
Level	4
Chrom-attention	1

The values of the parameters "level" and "chrom-attention" were chosen due to the values in the example video in the EMV algorithm website [16]. The $\omega_l = \frac{\pi}{2}\,[\frac{rad}{sec}]$ and $\omega_h = \frac{17}{30}\,[\frac{rad}{sec}]$ parameters were calculated using formula 6 and were chosen due to the RR value that were calculated in the respiration test. The α parameter was chosen due to the video and the graph that compared the RR in the EMV to the graph of the RR in the respiration test. The goal in the selection of the parameters values was to get the RR that would be as close as possible to the RR of the respiration test signals, within the assumption that the respiration test signal is a more accurate measurement for RR then the EMV.

2.2.6. RR Extraction from respiration test

The RR value presents the respiration rate that would have been achieved if the rate would have been based on the rate that was calculated between two adjacent peaks in the graph. In order to calculate this parameter, it was needed to calculate the duration time between two adjacent peaks and then calculate how many peaks would have been detected during 1 minute if the rate was constant. The time difference between two adjacent peaks in the signal was calculated using the built-in "findpeaks" MATLAB function and then each RR value was kept in a vector in order to display the RR graph as a function of time next to the respiration voltage graph.

2.2.7. Compression between the HR which extracted

The comparison between the two RR signals that were calculated by the different measurements was based on the same calculation which describe at 2.1.7 section.

3. RESULTS

The results for this experiment were calculated and displayed using the MATLAB software based on the algorithms, calculations and formulas described in the methods section.

3.1 Part 1

In part 1 of the experiment the examinee was seated on a chair while an ECG measurement device was measuring her heart activity and at the same time a camera recorder was filming her. The results of the ECG measurement were edited so the HR signal of the examinee displayed on graphs:

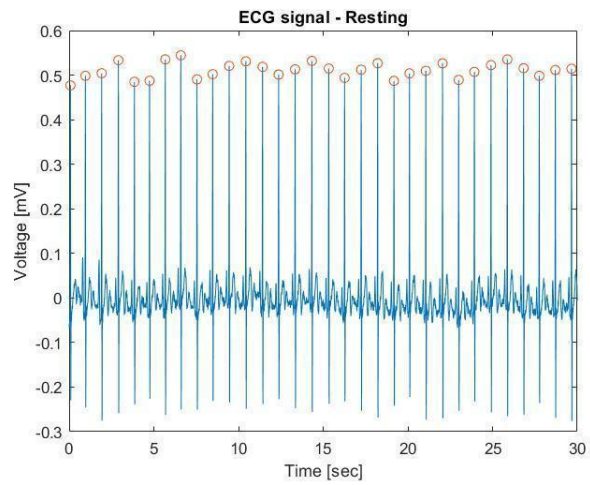

Figure 7: Graph of the voltage [mV] vs. time [sec] for the ECG signal at rest

On the graph at figure 7 above, a very moderate and consistent heart rate can be seen with the peaks of the heart beats marked in red circles. The peaks' voltage was almost constant around 0.5 mV and the distance between the peaks looks also very constant.

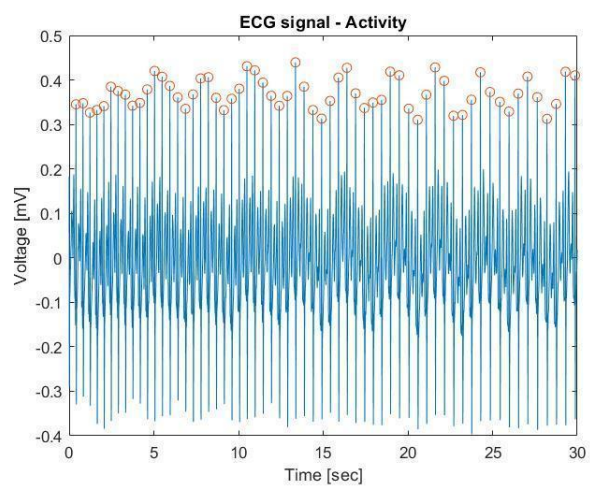

Figure 8: Graph of the voltage [mV] vs. time [sec] for the ECG signal at activity

The graph at figure 8 above is showing very frequent heart beats with voltage peaks around 0.4 m. It is also very noticeable that the distance between the peaks is pretty constant. On the other

hand, it can be seen that in the beginning of the recording, the peaks were at lower voltages and the variance of the voltage looked small, while as the recording continued, the peaks' voltage rises and the variance between them is higher.

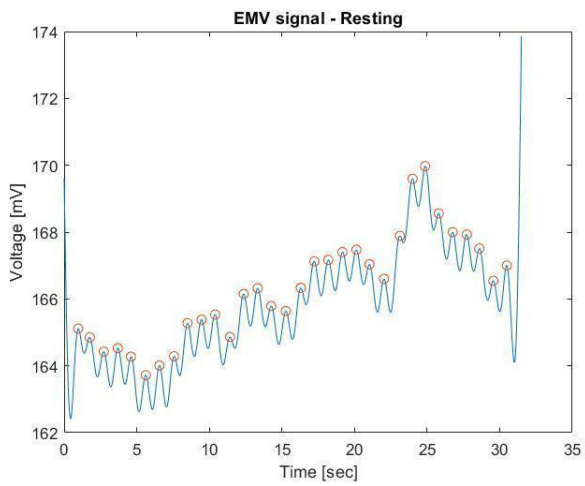

Figure 9: Graph of the voltage [mV] vs. time [sec] for the EMV signal at rest

In the graph at figure 9 of the EMV signal at rest, it can be seen that the voltage had an overall increasing trend and the distance between the peaks looks closely constant even though the voltage is changing.

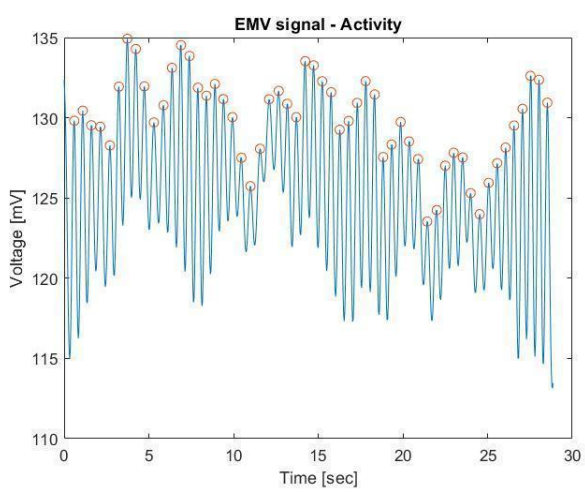

Figure 10: Graph of the voltage [mV] vs. time [sec] for the EMV signal at activity

The graph at figure 10 of the EMV signal after exercise presents very frequent peaks with a closely constant distance between the peaks, although the voltage is ever changing and there is not a certain trend of the peaks voltages on this graph.

30 frames of the original and magnified videos recorded at rest and after exercise were extracted and are displayed on the next pages, in order to compare the results of the magnification using the EMV algorithm over the frames of the videos.

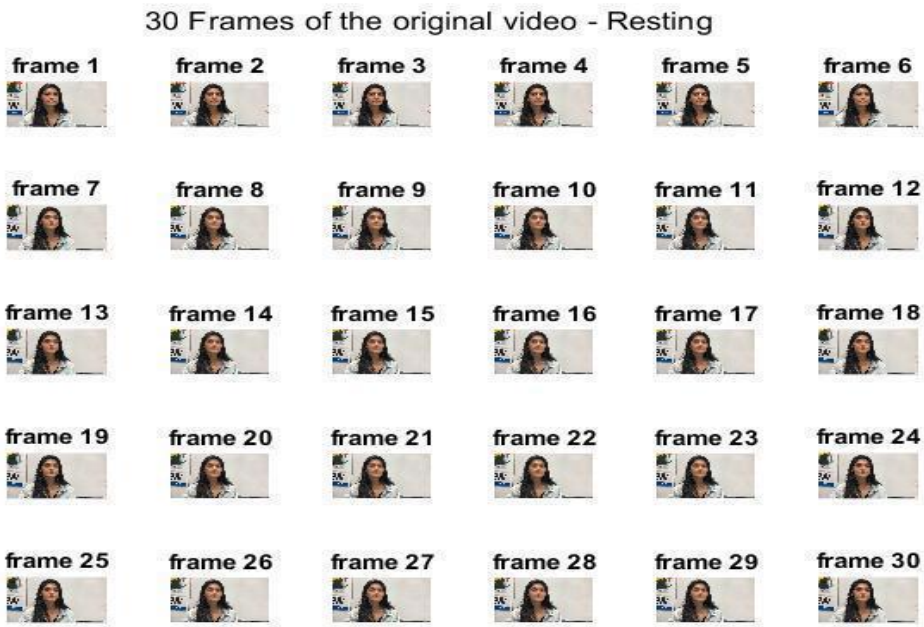

Figure 11: 30 frames of the original video – Resting

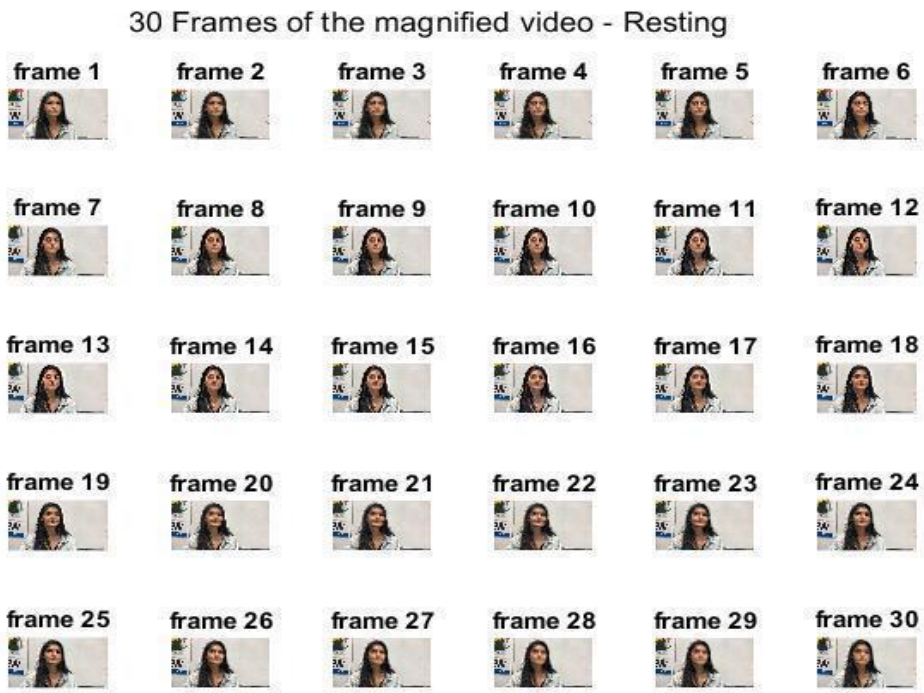

Figure 12: 30 frames of the magnified video – Resting

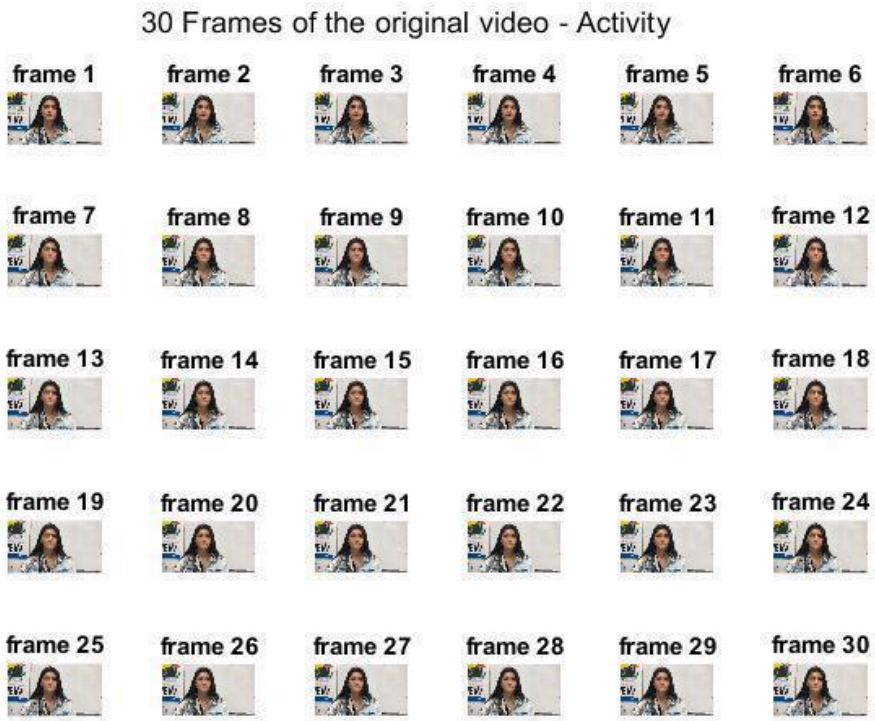

Figure 13: 30 frames of the original video – Activity

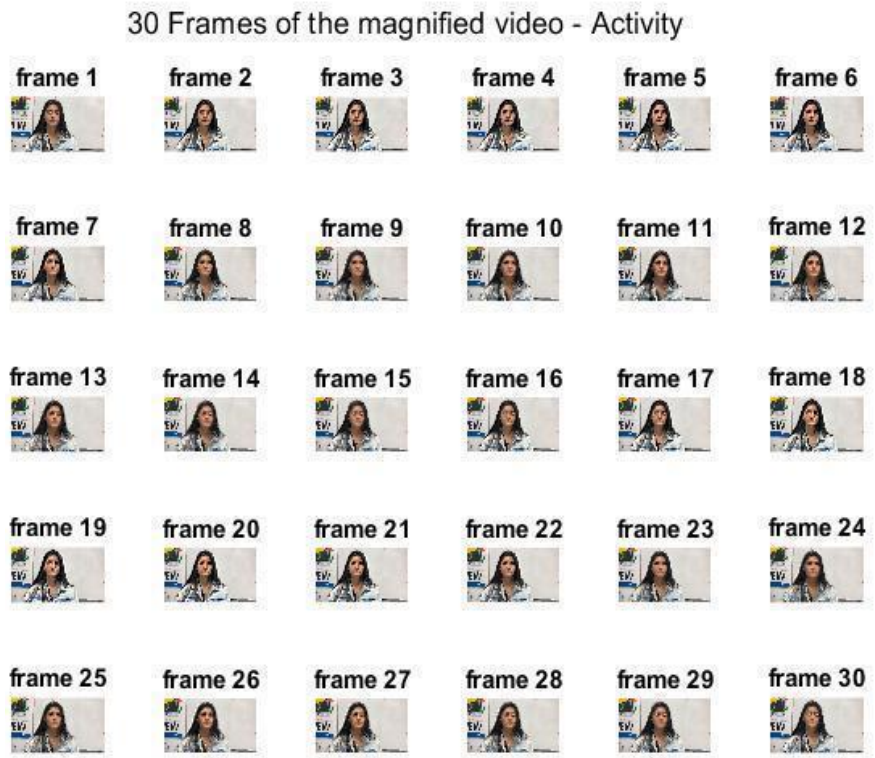

Figure 14: 30 frames of the magnified video – Activity

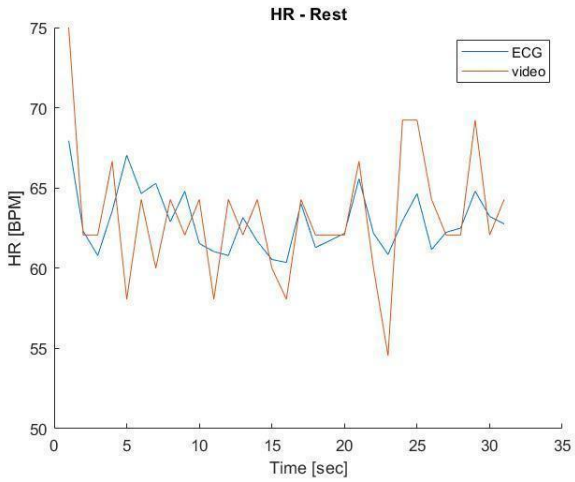

Figure 15: Graph of the HR [bpm] vs. time [sec] for the ECG and the EMV signals at rest

As it can be seen in the graph at figure 15 above, the HR that was calculated by the ECG at rest had a very similar trend as the HR that was calculated from the magnified video by the EMV method at rest.

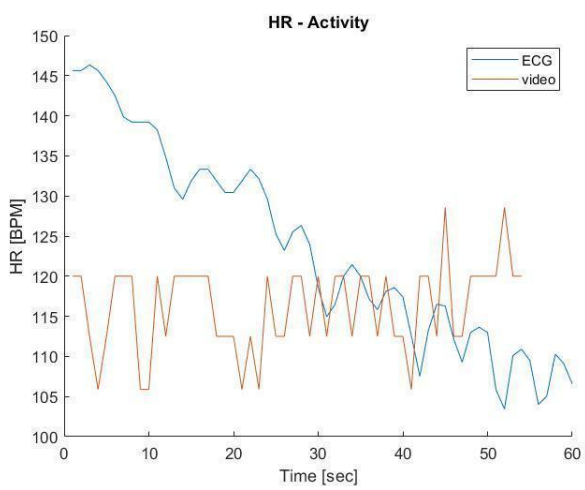

Figure 16: Graph of the HR [bpm] vs. time [sec] for the ECG and the EMV signals at activity

From the graph at figure 16 it is clearly noticeable that the HR of the examinee after the exercise decreased pretty fast during the ECG record. On the contrary, this decrease in the HR is not recognized in the video record, as the HR in this part seems like keeping on a moderate change around an almost constant average. Besides, at the first seconds until 30 seconds from the beginning of the recording, the HR in that extracted from the video was lower than the HR that was calculated by the ECG. But after 40 seconds from the beginning of the recording, the HR in the video was higher than the HR in the ECG.

Table 3: Comparing the average HR, variance HR, relative error (formula 2) and t-test results for the ECG and EMV measurements at rest and activity (with the chosen parameters)

Measures	rest	Active
Average ECG HR [BPM]	62.919	123.403
Average EMV HR [BPM]	63.217	116.249
Relative Error [%]	0.005	0.058
Variance ECG HR [BPM]	3.722	155.691
Variance EMV HR [BPM]	15.988	30.103
t-test	0	1

From the statistical calculations that are shown in the table 3 above, the relative error between the average HR in the ECG and the EMV at rest was 0.47% and so, beneath 5%. On the other hand, the relative error after exercise was 5.8% and so, a little bit above 5%. In addition, the overall average in the ECG at rest was lower than the HR average in the EMV, while the HR average after activity was higher than the average in the EMV. Also, the standard deviation of the HR in the ECG at rest was lower than the variance of the HR in the EMV, while after activity the variance in the ECG was much higher than the variance in the EMV. According to the results of the t-test the difference in the HR in the ECG signal compared to the HR in the WMV signal is that at rest there is no significant difference while, between the activity signals there is a significant difference.

The results applying $\omega_l = 1.8\ [Hz]$ and $\omega_h = 2.1\ [Hz]$ in both resting and activity recordings:

Table 4: Comparing the average HR, variance HR, relative error (formula 2) and t-test results for the ECG and EMV measurements at rest and activity (with the set parameters)

Measures	rest	Active
Average ECG HR [BPM]	62.919	123.403
Average EMV HR [BPM]	27.717	114.889
Relative Error [%]	0.560	0.069
Variance ECG HR [BPM]	3.722	155.691
Variance EMV HR [BPM]	274.352	91.780
t-test	1	1

As shown in the table 4 above the relative error in both resting and activity average HR between ECG measurement and EMV method were above 5% with 55.9% for the resting error and 6.9% for the activity error. In addition, the average HR in both resting and activity in ECG was higher than those extracted from the video with the EMV. On the other hand, the variance from the average HR at resting in the EMV was much higher than the variance in the ECG, while the opposite was noticed at the activity measurements as the variance in the ECG was much

higher than the variance in the EMV. In addition, according to the t-test there was not a significant difference between the HR at rest in the ECG and EMV signals, while between the activity signals there was a significant difference in the HR.

When comparing the HR average of ECG and EMV with the parameters that were set due to the explanations described at the methods part, to the ones with the parameters $\omega_l = 1.8\ [Hz]$ and $\omega_h = 2.1\ [Hz]$ the relative errors that achieved are:

Table 5: Comparing the average HR in EMV in the chosen and set parameters with the relative error between the results

	Rest	active
EMV average - chosen parameters [BPM]	63.2174	116.2488
EMV average – set parameters [BPM]	27.7171	114.8886
relative error [%]	0.562	0.012

Part 2

The Respiration Rate (RR) calculated, as described on formula 8 in order to expose the changing in the breathing rate over time. The results of the respiration test were edited so the RR signal of the examinee displayed on graphs.

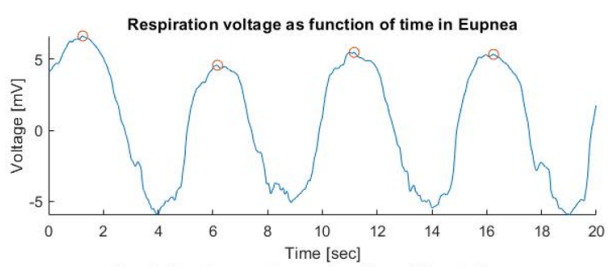

Figure 17: Voltage [mV] signal vs time of respiration test in eupnea

As we can see in the graph 17 above, the respiratory rate looks quite constant and moderate as expected in eupnea as the distance between the peaks is quite constant and the voltage of the peaks also seems constant around 5mV.

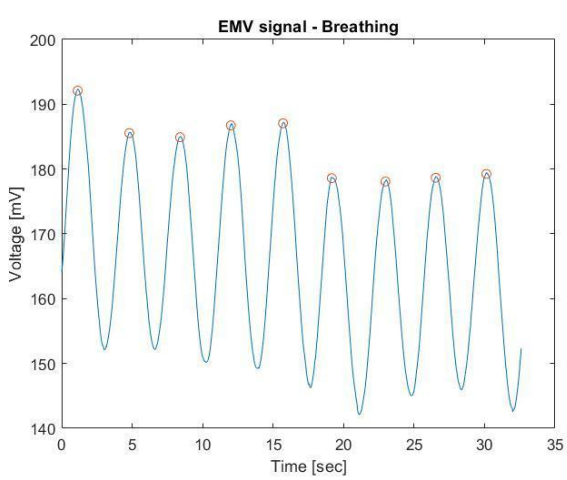

Figure 18: Voltage [mV] signal vs time of EMV in eupnea

From the graph in figure 18 above, a moderate decreasing trend can be seen, with a quite constant distance between the peaks. Even though there seems to be a decrease in the voltage over time, the overall voltage is around 180-190 mV.

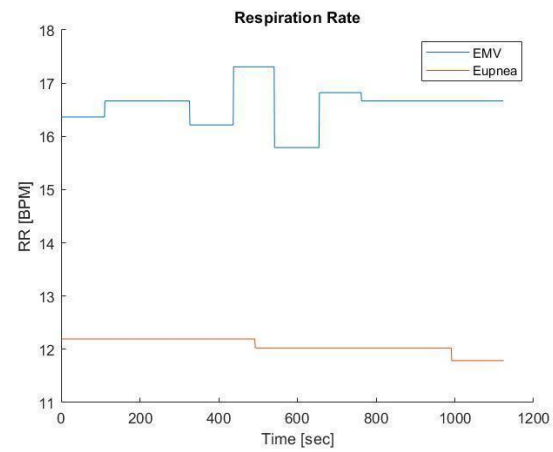

Figure 19: RR [BPM] as function of time [sec] of EMV test and respiration test

The RR values which were obtained during the respiration test experiment are quality constant overtime. The RR values which were obtained during the EMV experiment are relatively close to each other but are not fixed overtime.

Table 6: Theoretical & Experiment results

	mean [BPM]	variance [BPM]
Experiment Value (EMV)	16.578	0.140
Experiment Value (respiration test)	12.070	0.017
Theoretical value	8-16 [15]	

It can seem that the RR value obtained from the EMV measurement is higher than the RR value obtained from the respiration test. In addition, the variance of EMV is higher than the variance of the respiration test.

Table 7: Errors between theoretical & experiment results (formula 2)

	Relative error [%]
Experiment Value (EMV) vs Experiment Value (respiration test)	37.34
Experiment Value (EMV) vs Theoretical value	3.613

Experiment Value (respiration test) vs Theoretical value	in the range

The error between the value obtained from the EMV experiment to the theoretical value was between the upper limit of the theoretical range (16 [BPM]) to the value obtained from the EMV.

In addition, a t-test was calculated between the two experimental values in order to further examine the match between the values. The value that was obtained is 1, meaning there is a significant difference.

4. Discussion & Conclusion

4.1. Part 1

From the results of part 1 of the experiment, several conclusions can be obtained. First, the frames that are extracted from the original video don't contain any noticeable changes, but from the frames obtained in the magnified video, a change in the colors of the images can be seen between the frames. This result is in alignment with the expectation since the magnified video with the right frequencies chosen is assumed to make changes in the video color in the rhythm of the HR.

Secondly, from the graphs in figures 7 and 8 of the ECG and EMV voltage over time, the heart beats trends support the expectations with a moderate and distant heart peaks at rest and very frequent and heart peaks after exercise. Despite that coherence to the expectations, the EMV and ECG trends did not match. On the EMV signal, an increasing trend was clearly seen for the resting graph, and an uncertain trend was observed for the activity graph. These results can be explained by the fact that ECG signals are more accurate as this measurement calculates the heart activity by electrodes that are connected to the examinee itself and are very sensitive specifically to the heart activity. On contrast, the EMV is not an accurate measurement device as it extracts the heart activity from the film recording which Beyond the errors in the transfer of information between different devices in addition to the manufacturer's errors and the need to decrease the place in the devices' memory, this measurement affected by additional noises like movement of the object, changes in light in the room and shadow that added to the image for example.

In addition, as the graphs in figures 15 and 16 that compared the HR at rest and after exercise express, there was a good match between the HR that measured from the ECG and the EMV, which can imply that there was less noise in the resting record, and it might be due to the fact that a very frequent signals are at high frequencies, and also noises are frequent at high frequencies, like small and frequent movement or video noises that caused by light for example, and thus the denoising process is more difficult. That is in contrast to less frequent noises like blinking that can be removed by a simple FIR filter such as the FIR BP filter that was created. In addition, it is likely that after exercising the examinee moved more and that also could insert some noises. In correspondence to the results showing on the graph 8 of the voltage vs. time, it can be interfered that the ECG measurement after exercise could recognize changes in voltage variance in that way that as the time passed, the variance in voltage increased even though the time distance between the peaks retained constant. This phenomenon comes together with the HR graph after activity in figure 8 which shows a decrease in HR over time. This result can be explained because during the recording the examinee was seated at rest and thus her body started to regulate its breathing to return to homeostasis. For that manner the body voltage changed as the HR decreased to maintain a constant blood output.

In contrast, this phenomenon couldn't be recognized by the EMV measurement as the time difference between the heart's peaks did not change and thus a change in the face color kept on the same frequency. The results in the table 5 support the results from the graphs 7 and 8 as the difference between HR at rest was not significant with a very small relative error of 0.47% and under 5%, while the relative error after the exercise was significantly different according to the t-test with relative error of 5.8% which is above 5%. The high correspondence between the measurements of HR from the ECG and the EMV shows that the EMV succeeds in the evaluation of the HR of the examinee based on the changes in the examinee's change in face-color while at rest. On the other hand, the ability of the EMV to evaluate HR based on the change in face-color after activity is not accurate enough as the relative error crossed the 5% but only by 0.8% and thus maybe there is a need for better performance, or rougher requirements for the experiment procedure in order to improve the ability of EMV to evaluate HR based on the change in face-color.

In addition, when the parameters of $\omega_l = 1.8\ [Hz]$ and $\omega_h = 2.1\ [Hz]$ were set in the EMV there was not a significant change in the HR of the activity recording with relative error of 1.2% as the ωl and ωh that were chosen for the experiment were very close to these values with 1.8833 and 2.2167 respectively and thus, this result isn't surprising. Meanwhile, the parameters change affected dramatically on the resting part as the relative error is 56.2% and far above 5%. It should be mentioned that the incoherency for the resting part was expected as the chosen parameters for the resting part were not close to the parameters that were set with $\omega_l = 1\ [Hz]$ and $\omega_h = 1.6\ [Hz]$ respectively. In conclusion, it can be determined that the EMV method can well evaluate the HR at rest after some editing work using filters to reduce the noises. Meanwhile, the EMV ability to evaluate HR after exercise is not good enough, but also not too far, with only 0.8% cross from 5% relative error. Due to those results, it can be assumed that a few requirements over the experiment procedure can decrease the noises and enable good evaluation of HR also after exercise. Recommended requirements can be, firstly, the examinee should close his eyes in order to reduce blinking noises. In addition, a procedure that took in natural light can reduce light noises that resulted from artificial lightning conditions. Another requirement that can benefit the results can be to limit the transformation of the video from devices and formats in order to reduce the artificial noises that are added to

videos after transformations between different devices and different recording formats.

4.2. Part 2

When comparing the EMV breathing recording voltage over time at figure 18 to the respiratory test recording voltage over time it can be seen that that the frequency of the breaths in the EMV was higher than those obtained in the respiration as over 20 seconds only 4 peaks were recorded for the respiratory test while 6 peaks were recorded for the EMV measurement.

The RR values which were obtained from the EMV measurement are out of theoretical range, but the error which is calculated relative to the upper limit of the theoretical range is lower than 5% (table 6). Moreover, The RR value which was obtained from the respiration test measurement is in the theoretical range (table 6). Meaning both experiment values of RR are consistent with the theoretical range.

However, the RR value which was obtained from the EMV measurement is larger than the RR value which was obtained from the respiration test measurement. The error between them is higher than 5% (37.35%, table 7) and the value of the t-test has shown that there is a significant difference.

There are some possible reasons for that. First of all, the theoretical value itself is a range, and because of that it makes sense that the experiment values will change from time to time. In addition, the respiration tests were performed on the same subject and during the same breathing's simulation (rest), but on different days. The difference of the time each experiment was taken can influence the results dramatically due to the fact that the physical conditions of the examinee might have changed during the times that the measurements were taken, and also, this is possible that the breathing activity was influenced by the fact that the examinee was aware that her breathing is recording and thus she didn't performed her natural breathing activity.

In addition, the variance of EMV is higher than the variance of the respiration test. It can be explained by the fact that EMV is not an accurate measurement device as described before.

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http://people.csail.mit.edu/mrub/evm/#code

Appendix

Appendix A

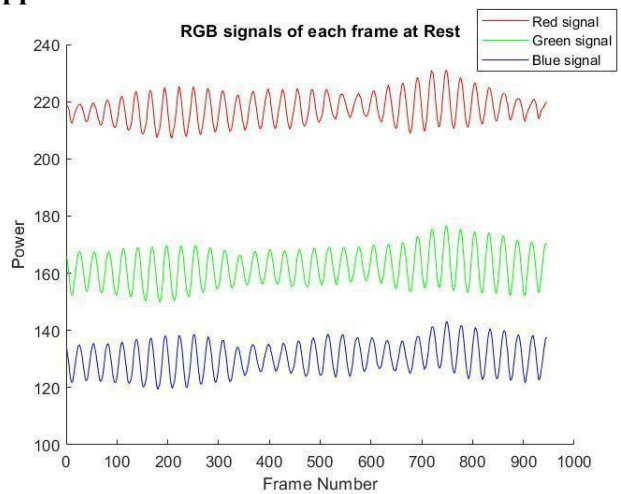

Figure 18: The Red, Green and Blue signals that were extracted from the video that captures the rest phase. The signal that was chosen was the Green signal.

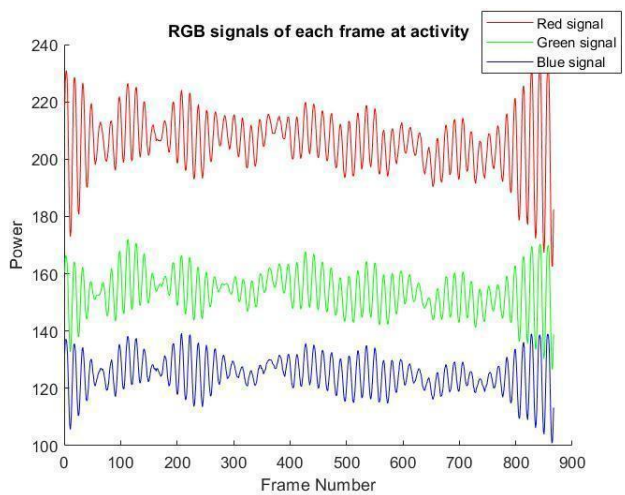

Figure 19: The Red, Green and Blue signals that were extracted from the video that captures the active phase. The signal that was chosen was the Blue signal.

Appendix B

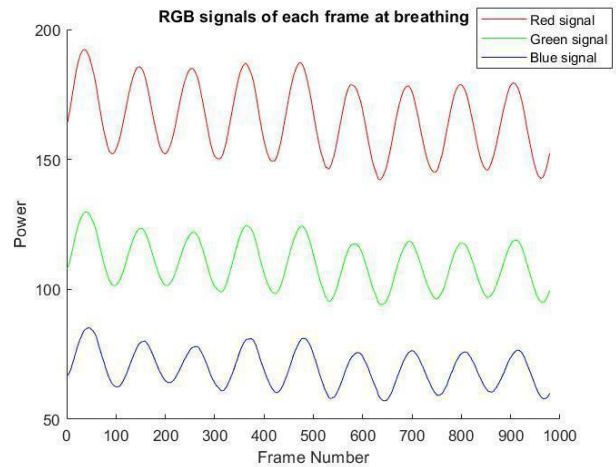

Figure 20: The Red, Green and Blue signals that were extracted from the video that captures the active phase. The signal that was chosen was the red signal.